Mika Be

SOFTWARE ENGINEERING | APPLIED AI

Piscataway, NJ | (201) 913-4835 | mika.be@rutgers.edu | mikabe1805.github.io | github.com/mikabe1805 | linkedin.com/in/mika-be

EDUCATION

Rutgers University, New Brunswick | B.S. Electrical and Computer Engineering, CS minor — Expected May 2028

GPA 3.838/4.000 | Dean's List, all four completed semesters

Honors 2nd Place Best Full Paper, IEEE ISEC 2026 | IEEE-HKN honor society | Bergen County Academies CS Award (1 of 3)

Coursework Data Structures; Digital Logic Design + Lab; Principles of EE I & II + Labs; Discrete Structures; Differential Equations

In progress Computer Architecture & Assembly + Lab; Linear Systems & Signals; Electronic Devices + Lab; Discrete Structures II

RESEARCH

Kuiper TTS, Rutgers ECE | Undergraduate Research Lead (PI: Prof. Dov Kruger) — Sep 2025-Present

Python | PyTorch Lightning | Piper/VITS | ONNX

- Led an 18-member team building offline text-to-speech to read scientific papers for blind and low-vision users
- Co-authored the 2nd Place Best Full Paper at IEEE ISEC 2026; designed its rules for reading equations aloud, tested on 46 equations from a 1971 antenna paper ("d q over d t" became "the derivative of q with respect to t")
- Built a GPU training pipeline and trained custom voices on 886 self-recorded clips; fixed a pitch bug in augmentation
- Created audio-cleanup, augmentation and phoneme-coverage tools the team ported to C++; wrote the voice-training guide

EXPERIENCE

this.is | Lead Software Developer (Internship) — Jul 2025-Aug 2026

React | TypeScript | Firebase | Cloud Functions | Google Places API

- Developed and deployed a startup's social place-discovery web app as primary engineer; mentored and managed 2 interns
- Built a Google Places recommendation engine favoring local businesses, with novelty tracking and geographic diversity
- Led a product pivot and built v2 as sole developer: a group planner that turns saved places into up to 3 suggestions

Schieber Research | Software Engineering Intern — Sep 2023-Jun 2024

- Analyzed competitors and wrote product documentation for this.is, a startup client building a discovery app
- Designed and built the app's first 3 prototypes in Node.js, React and Appwrite

Starbucks | Barista (part-time, alongside full-time coursework) — Jul 2023-Jun 2025

PROJECTS

Room of Days | Creator and Solo Developer, RPG-style habit app for iOS — Jun 2026-Present

Flutter | Dart | Swift (WidgetKit) | Firebase | TypeScript Cloud Functions | Codemagic CI

- Shipped to the App Store in Aug 2026 as sole designer and developer; 10+ active users, 5.0 stars (5 ratings)
- Designed the XP system: 6 life stats, difficulty ladders, streak freezes and a comeback bonus; no penalty for missed days
- Deployed a Firebase social backend: room sharing, an opt-in directory, server-side moderation and App Attest checks

Room Notes | Creator and Solo Developer, cross-device note-taking app (private beta) — Aug 2026-Present

Flutter | Dart | Objective-C++ | Drift/SQLite | Firestore | Cloud Functions

- Built a local-first notes app where iPad (Apple Pencil), phone (board photos) and laptop edit the same pages
- Traced a 1.5 s pen freeze after writing pauses to sync-echo handling and removed it, verified in a sync simulation
- Patched Flutter's iOS engine to deliver coalesced (between-frame) Apple Pencil input; shipped it via CI to TestFlight

Knowsy Breadboard Lab, Rutgers ECE | Team Lead, 3-person team (1 of 7 module leads) — May 2026-Present

JavaScript | SVG | Verilog

- Created a browser breadboard simulator for Digital Logic and EE II lab prep, matching real lab boards and 74LS pinouts
- Developed live breadboard-to-schematic rendering and debugging tools: logic probe, voltmeter and wiring-fault warnings
- Adapted course materials into 8 guided labs (90 auto-checked steps); built a nodal-analysis solver and Verilog export

Bio (Discord bot) | Creator and Developer — Sep 2025-Present

Node.js | SQLite | Railway

- Launched a Discord bot that connects members with shared interests through opt-in profiles; live in 2 servers

TECHNICAL SKILLS

Programming Python, TypeScript, JavaScript, Dart, C/C++, Java, SQL, Swift, Verilog

AI/ML PyTorch Lightning, ONNX, Piper/VITS TTS training, speech data prep (VAD, LUFS), AI coding agents (Claude Code, Codex)

Software Flutter, React, Node.js, Firebase (Firestore, Cloud Functions, App Check), Drift/SQLite, Git, CI/CD (Codemagic)

Hardware and EE tools Vivado, LTspice, MATLAB/Simulink, oscilloscopes, logic probes, soldering, Onshape CAD, 3D printing

Languages English (fluent), Hebrew (native), Spanish (conversational)